

TABLE VII
SHIFTS IN COLLEGE-EQUIVALENT LABOR DEMAND IMPLIED
BY CHANGES IN JOB TASKS, 1970–1998

	1. 1970– 1980 extensive margin	2. 1980– 1990 extensive margin	3. 1990– 1998 extensive margin	4. 1970– 1998 extensive margin	5. 1980– 1998 extensive margin	6. 1980– 1998 intensive margin	7. 1980– 1998 extensive + intensive
A. $10 \times$ observed annual changes in DOT task measures (percentile changes relative to 1960 task distribution)							
Nonroutine analytic	3.02	2.97	3.12	3.04	3.05	−0.39	2.67
Nonroutine interactive	4.68	5.31	4.48	4.84	4.85	0.58	5.43
Routine cognitive	−0.14	−3.48	−4.88	−3.03	−4.26	−2.76	−7.02
Routine manual	1.63	−1.47	−3.88	−1.44	−2.81	0.74	−2.07
B. $10 \times$ predicted annual changes in DOT task measures (percentile changes relative to 1960 task distribution)							
	NIPA computer input measure				CPS computer use measure		
Nonroutine analytic	0.84	1.35	2.30	1.55	2.56	0.54	3.10
Nonroutine interactive	1.47	2.36	4.01	2.70	3.16	1.04	4.20
Routine cognitive	−1.05	−1.68	−2.86	−1.92	−3.14	−3.32	−6.46
Routine manual	−1.15	−1.86	−3.15	−2.12	−2.84	0.32	−2.52
C. $10 \times$ predicted annual changes in college-equivalent share of employment in percentage points, due to observed task shifts (panel A)							
Nonroutine tasks	1.53	1.40	1.63	1.51	1.53	−0.36	0.83
Routine tasks	−0.20	0.66	1.17	0.50	0.97	0.27	0.87
All tasks	1.33	2.06	2.80	2.01	2.49	−0.09	2.40

Panel A: Observed extensive margin task shifts are defined as the change in economywide input of each task (in percentiles of the 1960 task distribution) estimated using DOT 1977 occupational task measures applied to Census and CPS samples for 1970 to 1998 and summarized in Table II. Intensive margin shifts are measured as change in mean of DOT occupational task input (measured in centiles of the 1977 task distribution) between 1977 and 1991 DOT revisions, using the 1980 and 1998 occupational distributions of employment from the CPS MORG samples. See Table I and Appendix 1 for definitions and examples of task variables.

Panel B: Predicted task changes are calculated as the weighted mean of the NIPA computer investment measure or CPS computer use measure (as noted) multiplied by the coefficient from a regression of changes in industry or occupation task input on the relevant computer measure. NIPA coefficient estimates correspond to specification (1) of Table IV. CPS extensive margin computer task estimates correspond to Panel A of Table V. CPS intensive margin computer task estimates correspond to specification (1) of Table VI. (A negligible interaction term is ignored.)